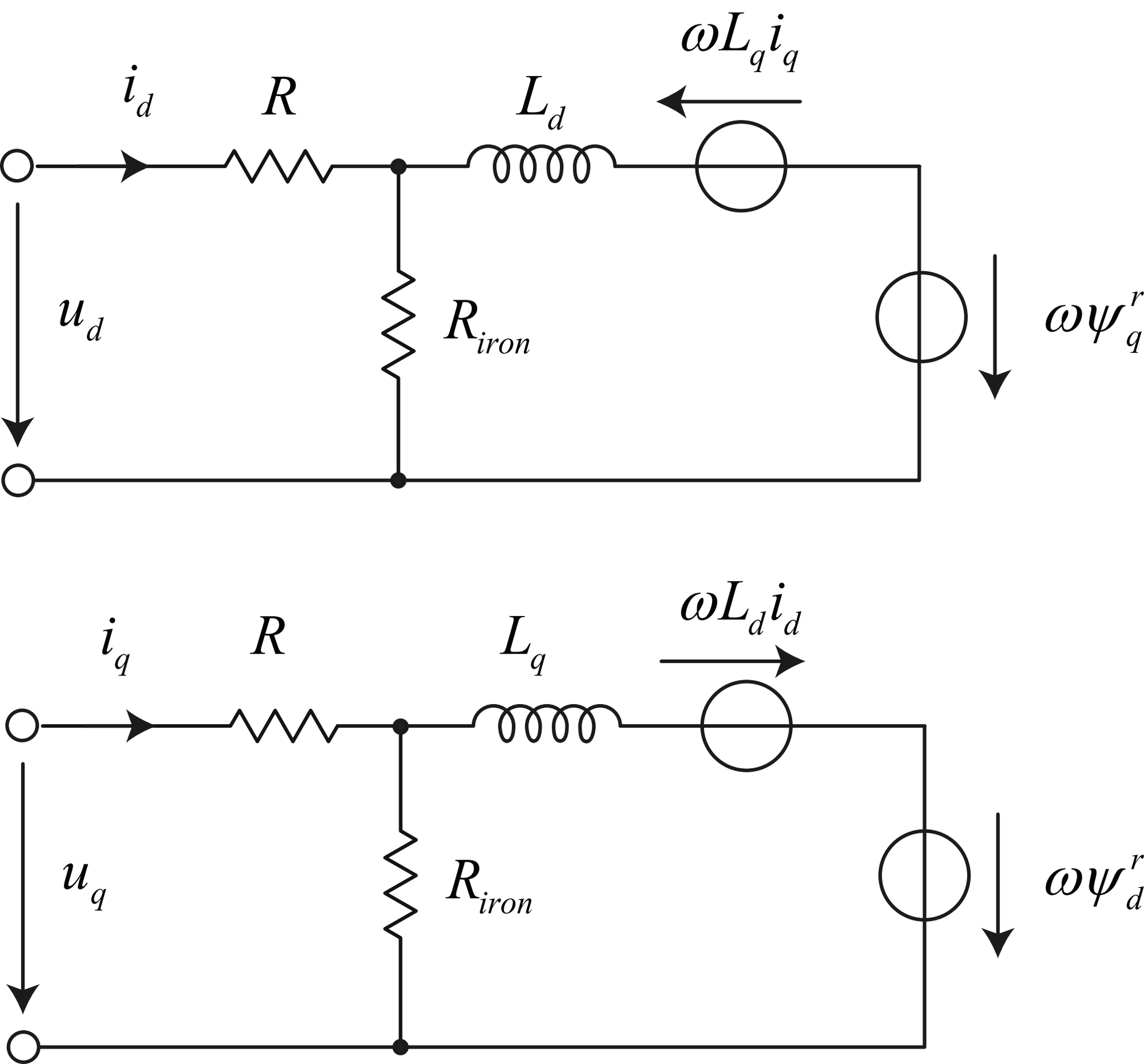
$\omega L_q i_q$
i_d
R
L_d
u_d
R_{iron}
$\omega \psi_q^r$
$\omega L_d i_d$
i_q
R
L_q
u_q
R_{iron}
$\omega \psi_d^r$